THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The distance of the point $(-8, -15)$ from the origin is: 1

(a) 8 units **(b)** 18 units **(c)** 23 units **(d)** 17 units

2. The distance between $A(5, 1)$ and $B(-7, 6)$ is: 1

(a) 14 units **(b)** 15 units **(c)** 12 units **(d)** 13 units

3. The point $(0, -4)$ lies: 1

(a) Quadrant III **(b)** on the y-axis

(c) Quadrant I **(d)** Quadrant II

4. The perimeter of a square of side x cm is a linear polynomial. For side 2.5 cm it is: 1

(a) 12.5 cm **(b)** 10 cm **(c)** 6.25 cm **(d)** 5 cm

5. The value of the linear polynomial $-2x - 1$ at $x = 5$ is: 1

(a) -13 **(b)** -11 **(c)** -9 **(d)** 11

6. The number π is: 1

(a) not a real number **(b)** rational

(c) neither rational nor irrational **(d)** irrational

7. The value of $0 - (-14)$ is: 1

(a) -14 **(b)** 16 **(c)** 14 **(d)** 18

8. The number $0.35\overline{7}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{161}{450}$ **(b)** $\frac{450}{161}$ **(c)** $\frac{9}{25}$ **(d)** $\frac{103}{225}$

9. $(3s + 5t)(3s - 5t)$ equals: 1

(a) $9s^2 - 30st + 25t^2$ **(b)** $9s^2 - 25t^2$ **(c)** $9s^2 + 30st + 25t^2$ **(d)** $9s^2 - 25t^2 + 4$

10. Expand $(4x + 3)^2$: 1

(a) $16x^2 + 9$ **(b)** $16x^2 + 12x + 9$ **(c)** $16x^2 + 24x + 9$ **(d)** $16x^2 + 24x - 9$

11. Of two chords of a circle, the one nearer to the centre is: 1

(a) equal in length **(b)** shorter

(c) longer **(d)** cannot be compared

12. An arc subtends an angle of 130° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 130° **(b)** 260° **(c)** 115° **(d)** 65°

13. An arc subtends 41° at a point on the circle. The angle subtended by the same arc at the centre is: 1

(a) 82° **(b)** 139° **(c)** 41° **(d)** 42°

14. By Heron's formula, the area of a triangle with sides 9 cm, 10 cm and 17 cm is: 1

(a) 90 cm^2 **(b)** 36 cm^2 **(c)** 42 cm^2 **(d)** 30 cm^2

15. The perimeter of a circle is 44 cm. Taking $\pi = \frac{22}{7}$, its radius is: 1

(a) 14 cm **(b)** 7 cm **(c)** 8 cm **(d)** 3.5 cm

16. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 3.5 cm is: 1

(a) 38.5 cm **(b)** 44 cm **(c)** 22 cm **(d)** 11 cm

17. The probability of a certain event is: 1

(a) 100 **(b)** 0.5 **(c)** 1 **(d)** 0

18. A sample shows a probability of $\frac{3}{20}$ that a student likes cricket. In a school of 800 students the expected number who like cricket is: 1

(a) 200 **(b)** 120 **(c)** 220 **(d)** 60

19. The sequence $1, 3, 6, 10, 15, \ldots$ (triangular numbers) is: 1

(a) neither an AP nor a GP **(b)** an AP with d = 5

(c) an AP with d = 2 **(d)** a GP with r = 3

20. The 6th term of the GP with first term 3 and common ratio 2 is: 1

(a) 192 **(b)** 96 **(c)** 97 **(d)** 95

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Find the coordinates of the midpoint of the segment joining $P(4,\ 6)$ and $Q(10,\ -2)$. 2

22. A linear pattern begins $2,\ 7,\ 12, \ldots$ Write the expression for its nth term and find the 12th term. 2

23. The temperature in Ladakh is 4 °C at noon and drops by 15 °C by midnight. Find the midnight temperature. 2

24. Factorise $16y^2 - 24y + 9$. 2

25. A cyclic quadrilateral has angles $\angle A = 80^\circ, \angle B = 110^\circ, \angle C = 100^\circ, \angle D = 70^\circ$. Can it be drawn? Why? 2

26. Find, correct to 3 significant figures, the circumference of a circle of radius 10 cm ($\pi = \frac{22}{7}$). 2

27. A die is rolled 50 times and shows a 4 eight times. Find the experimental probability of a 4 and compare it with the theoretical probability. 2

28. Find the first five terms of $t_n = 2 - 5n$. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Show that the point $(0,\ -1)$ is equidistant from $(-4,\ 3)$ and $(4,\ 3)$. Hence name the type of triangle formed by the three points, with reason. 3

30. Write the coordinates of three points on the line $y = 3x - 2$ and state where it cuts the y-axis. 3

31. Show that $\sqrt{2}$ can be located on the number line and describe the construction. 3

32. Prove that the difference of the squares of two consecutive integers equals their sum. 3

33. A parallelogram has base 20 cm, height 9 cm and one side 15 cm. Find its area and the distance between its other pair of parallel sides. 3

34. A number is chosen at random from 1 to 25. Find the probability that it is 3
(i) a multiple of 5
(ii) a prime number.

35. A chord of a circle of radius r is at distance d from the centre. Derive the chord length. Hence find it for $r = 10$ and $d = 6$. 3

OR

ABCD is cyclic, $\angle BAC = 30^\circ$ and $\angle BCD = 100^\circ$. Find $\angle CBD$. 3

36. Find the 10th, 17th and 80th triangular numbers. 3

OR

Write the first ten terms of the Virahānka–Fibonacci sequence ($V_1 = 1, V_2 = 2$). 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Simplify $\left(\frac{1}{4} - \frac{5}{12}\right) \div \frac{2}{3}$. 5
(ii) Find the sum and the product of $\frac{3}{5}$ and $\frac{7}{10}$.

OR

(i) Find the rational number x such that $x\left(\frac{1}{3} + \frac{1}{6}\right) = \frac{9}{2}$. 5
(ii) Verify $\frac{1}{2}\left(\frac{3}{4} + \frac{1}{3}\right) = \frac{1}{2} \times \frac{3}{4} + \frac{1}{2} \times \frac{1}{3}$.

38. A cyclic quadrilateral has sides 5, 5, 12, 12 (in order). 5
(i) Show that its diagonal is a diameter and find the radius.
(ii) Find its area.
(iii) Explain why this quadrilateral is not a rectangle.

OR

Case study: A round table has centre O and radius 60 cm. Three diners sit at A, B, C on its edge with $\angle BAC = 40^\circ$. 5
(i) Find $\angle BOC$.
(ii) Find $\angle OBC$.
(iii) A fourth diner sits at D on the same arc as A; find $\angle BDC$.
(iv) If M is the midpoint of BC and $OM = 36$ cm, find BC.

39. Prove, using the inscribed and circumscribed regular hexagons of a circle of radius r, that $3 < \pi < 2\sqrt{3}$. Why is Archimedes' extension to 96 sides useful? 5

OR

(i) List the integer-sided rectangles of perimeter 40 units and find their areas. 5
(ii) Which has the greatest area and which the least?
(iii) What do you notice about the shape of the rectangle of greatest area?

40. **Case study:** Meera saves ₹100 in the first month and ₹50 more each month than in the previous month. 5

(i) Write her savings for the first three months.

(ii) Find the nth term.

(iii) Find her saving in the 12th month and her total in a year.

(iv) In which month does she save ₹1,000?

OR

A sequence has $t_1 = -5$ and $t_{n+1} = t_n + 3$. 5

(i) Write the first five terms.

(ii) Find the explicit rule.

(iii) Is 52 a term?

(iv) Find the sum of the first 20 terms.

END OF QUESTION PAPER **Total: 80 Marks**

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